



St. Francis Institute of Technology

Department of Computer Engineering

Criminal Code Summarization and Outcome Prediction Using InLegalBERT and Bharatiya Nyaya Sanhita Mapping

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Content

- Introduction
- Existing System
- Literature Review
- Problem definition
- Proposed Solution
- Flowchart
- Experimental Setup
- Output
- Conclusion
- References



Introduction

- The Bharatiya Nyaya Sanhita (BNS) Act introduces a modernized legal framework, replacing the long-standing Indian Penal Code (IPC), which has faced challenges due to outdated language and complex classifications.
- The BNS Act simplifies these classifications, aiming to enhance accessibility, efficiency, and relevance in criminal law processes.
- The transition to the BNS framework presents challenges for legal professionals who rely on IPC-based tools, as existing systems are not compatible with the new framework.



Introduction

- There is a need for BNS-specific tools and datasets to assist with tasks such as legal document summarization, section mapping, and case outcome prediction.
- This project aims to bridge the gap by creating a structured dataset and a rule-based mapping strategy to align IPC sections with their BNS equivalents, improving legal document processing under the new BNS framework.



Existing System

- Existing systems are designed for IPC-based analysis and lack the structural adjustments needed for the BNS Act, resulting in a lack of BNS-compatible datasets.
- There is no systematic mapping of IPC sections to their BNS equivalents, making it difficult to interpret or apply historical data within the new framework.
- Current legal document summarization models are tailored to IPC-based cases, failing to align with the simplified classifications of BNS.



Literature Review

- Paul, et al. [1] retrained LegalBERT and CaseLawBERT on Indian legal data, finding that InLegalBERT showed significant improvements in legal document summarization, especially in understanding legal terminology and context.
- S. Sharma et al. [2] explores using InLegalBERT for legal document summarization. Evaluated against Legal Pegasus, T5 base, BART, and BERT using ROUGE-L F1 scores, InLegalBERT achieves the best performance for Indian legal texts.



Literature Review

- Rani, U. and Bidhan, K. [3] provides a comparative analysis of various extractive summarization techniques, including TextRank, TF-IDF, and LDA, evaluating their effectiveness in condensing textual content, especially in contexts requiring concise and relevant summaries .



Problem Definition

The shift from IPC to BNS creates challenges in legal document analysis due to the lack of a BNS-specific dataset and tools. Existing systems, designed for IPC, cannot efficiently handle the new BNS structure, hindering tasks like summarization, section mapping, and outcome prediction. This highlights the need for tailored tools and datasets to support legal analysis under the BNS framework.



Proposed Solution

1. Create synthetic BNS dataset mapping IPC to BNS
2. Use InLegalBERT for generating embeddings for extractive summarization
3. Fine-tune BART for abstractive summarization
4. Present case findings in detail
5. Rule-based IPC to BNS mapping



Proposed Solution

1. Create synthetic BNS dataset mapping IPC to BNS

- Since no official BNS dataset exists, a synthetic dataset was created.
- IPC sections were manually mapped to their corresponding BNS sections.
- This dataset serves as the foundation for training and testing the system on BNS cases.
- Ensures structured analysis and continuity between IPC and BNS cases.



Proposed Solution

2. Use InLegalBERT for generating embeddings for extractive summarization

- InLegalBERT, a legal-domain pre-trained model, is used for sentence embedding.
- Each sentence from the legal document is transformed into a dense vector.
- Sentence similarity is computed using embeddings to select key sentences.
- Helps extract the most relevant parts of the document while preserving legal meaning.



Proposed Solution

3. Fine-tune BART for abstractive summarization

- A fine-tuned BART model (facebook/bart-large) is used for summarization.
- Extracted sentences are further processed to generate human-like, rephrased summaries.
- BART handles complex legal terminologies and produces coherent, fluent summaries.
- This improves readability without losing important legal facts.



Proposed Solution

4. Present case findings in detail

- After summarization, detailed case analysis is performed.
- Important aspects like case background, evidence, legal sections, and outcomes are extracted.
- Summarization is complemented with IPC and BNS code references.
- Helps legal professionals understand the case in a concise but informative manner.



Proposed Solution

5. Rule-based IPC to BNS mapping

- A rule-based system identifies IPC sections in the text using regex and NER (Named Entity Recognition).
- It then maps them automatically to the corresponding BNS sections.
- The mapping ensures that legal documents can be interpreted accurately under the new BNS framework.
- This strategy bridges the gap between historical IPC cases and the new legal structure.



Flowchart

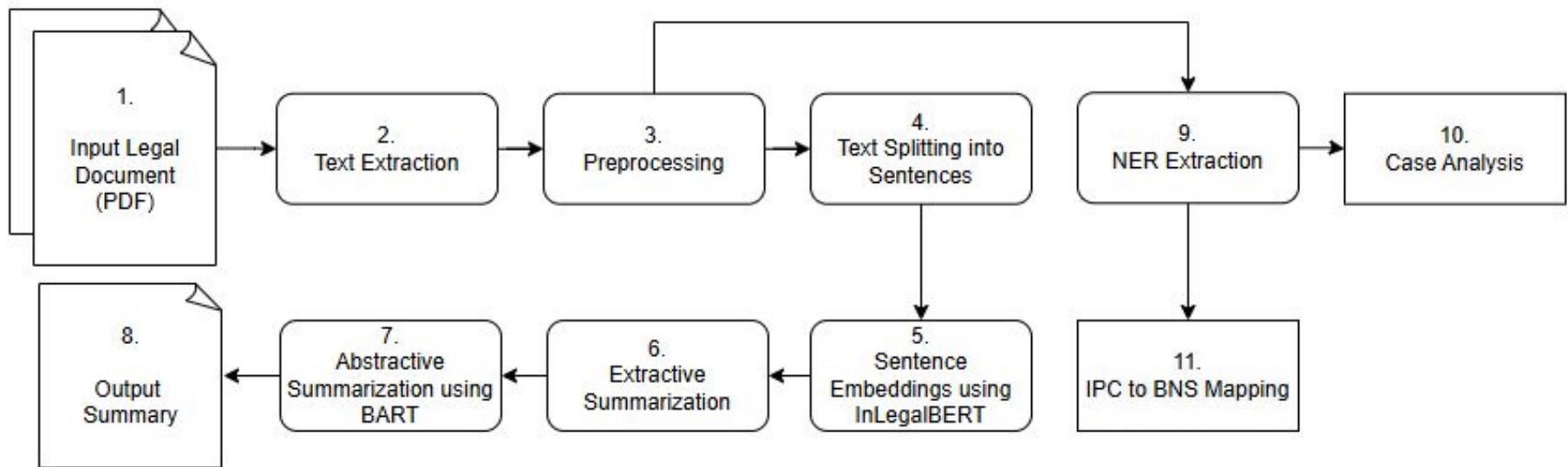


fig. (1) Flowchart for document summarization and sections mapping



Flowchart

1. **Input Legal Document**
2. **Text Extraction from the document**
3. **Preprocessing (cleaning, formatting)**
4. **Text Splitting into sentences**
5. **Sentence Embeddings using InLegalBERT**
6. **Extractive Summarization**
7. **Abstractive Summarization using BART**
8. **Output Final Summary**



Flowchart

9. **NER Extraction (find legal entities like dates, sections)**
10. **Case Analysis (case details, outcomes)**
11. **IPC to BNS Mapping (update legal sections)**



Experimental Setup

- **Dataset:** 7100 Indian Supreme Court judgments
- **Models Used:**
 - InLegalBERT (Extractive Summarization)
 - Facebook/BART- (Abstractive Summarization)
- **Evaluation Metrics:** ROUGE, BLEU, METEOR, BERTScore
- **Hardware:** Windows 11, 16GB RAM, Ryzen 6800H, RTX 3050 GPU



Experimental Setup

- **Dataset :**
 - **Data Source:** The dataset was created from 7.1k Indian Supreme Court case documents.
 - **Data Splits:** The data is divided into two splits: 'train' (7k docs) and 'test' (100 docs).
 - **Data Instance:** An example from 'train' looks as follows:

```
{  
  "id": "2858"  
  "num_doc_tokens": 4867  
  "num_summ_tokens": 708  
  "document": ["Appeal No. 36 of 1967.",  
               "Appeal by special leave...", ...]  
  "summary": ["The appellant executed a usufructuary mortgage ...",  
              "In 1949, the mortgagee left for", ...]  
}
```

fig. (2) Dataset Instance Example



Experimental Setup

- **Models Used:**

- InLegalBERT (Extractive Summarization)

- This model is used to generate legal sentence embeddings for extractive summary.
 - The model has the configuration:
 - 12 hidden layers
 - 768 hidden dimensionality
 - 12 attention heads
 - ~110M parameters.

<https://huggingface.co/law-ai/InLegalBERT>



Experimental Setup

- **Models Used:**

- Facebook/BART- (Abstractive Summarization)

- This model is a large variant of BART and is effective for text generation
 - The model has the configuration:
 - 12 hidden layers in the encoder and decoder
 - 1024 hidden dimensionality
 - 16 attention heads
 - ~406M parameters.

<https://huggingface.co/facebook/bart-large>



Experimental Setup

- **Evaluation Metrics**

- ROUGE (Recall-Oriented Understudy for Gisting Evaluation)
 - ROUGE-1: Measures unigram (single word) overlap between generated and reference summaries.
 - ROUGE-2: Measures bigram (two consecutive words) overlap.
 - ROUGE-L: Measures the longest common subsequence of words, reflecting sentence-level structure.



Experimental Setup

- **Evaluation Metrics**

- BLEU (Bilingual Evaluation Understudy)

- Checks how many matching n-grams (phrases) exist between generated and reference summaries.

- METEOR (Metric for Evaluation of Translation with Explicit ORdering)

- Considers synonym matching, word order, and stemming for better fluency evaluation.

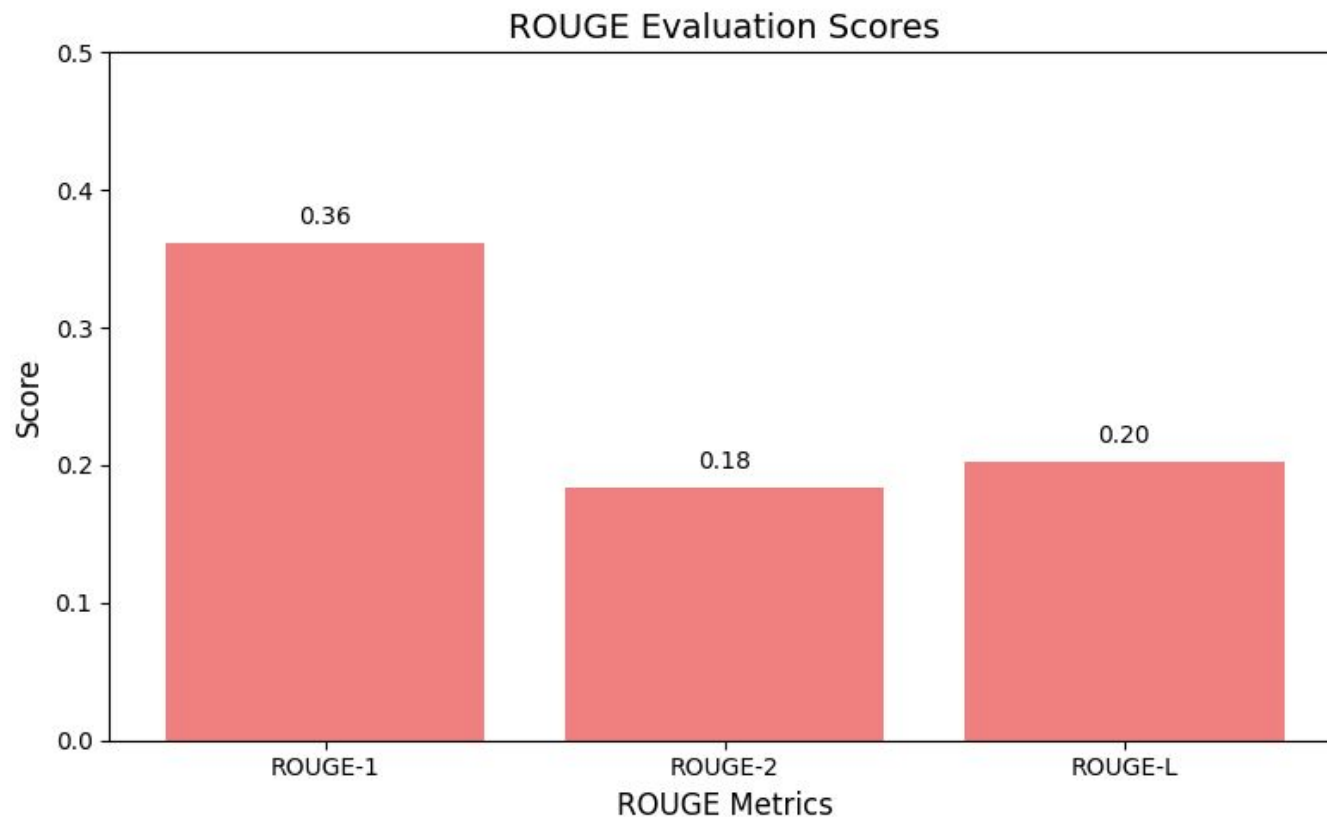
- BERTScore

- Uses contextual word embeddings to measure how semantically similar the summaries are.



Output

1. Fine-Tuned BART model evaluation results



Output

1. Fine-Tuned BART model evaluation results

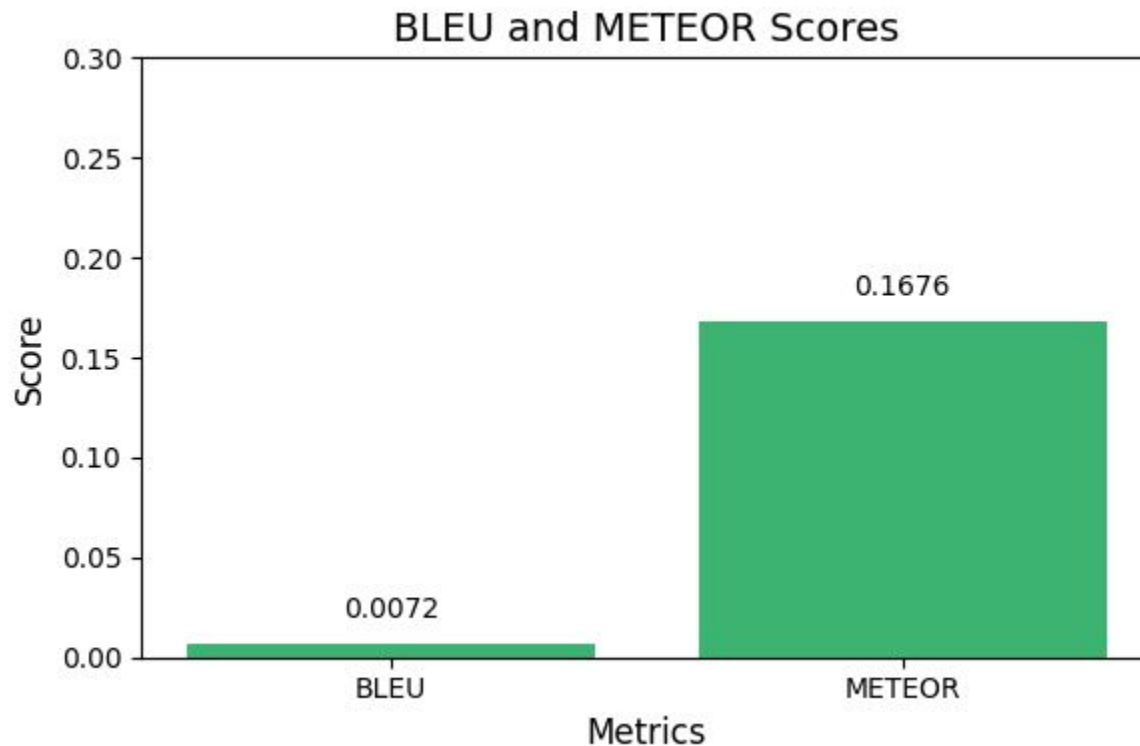
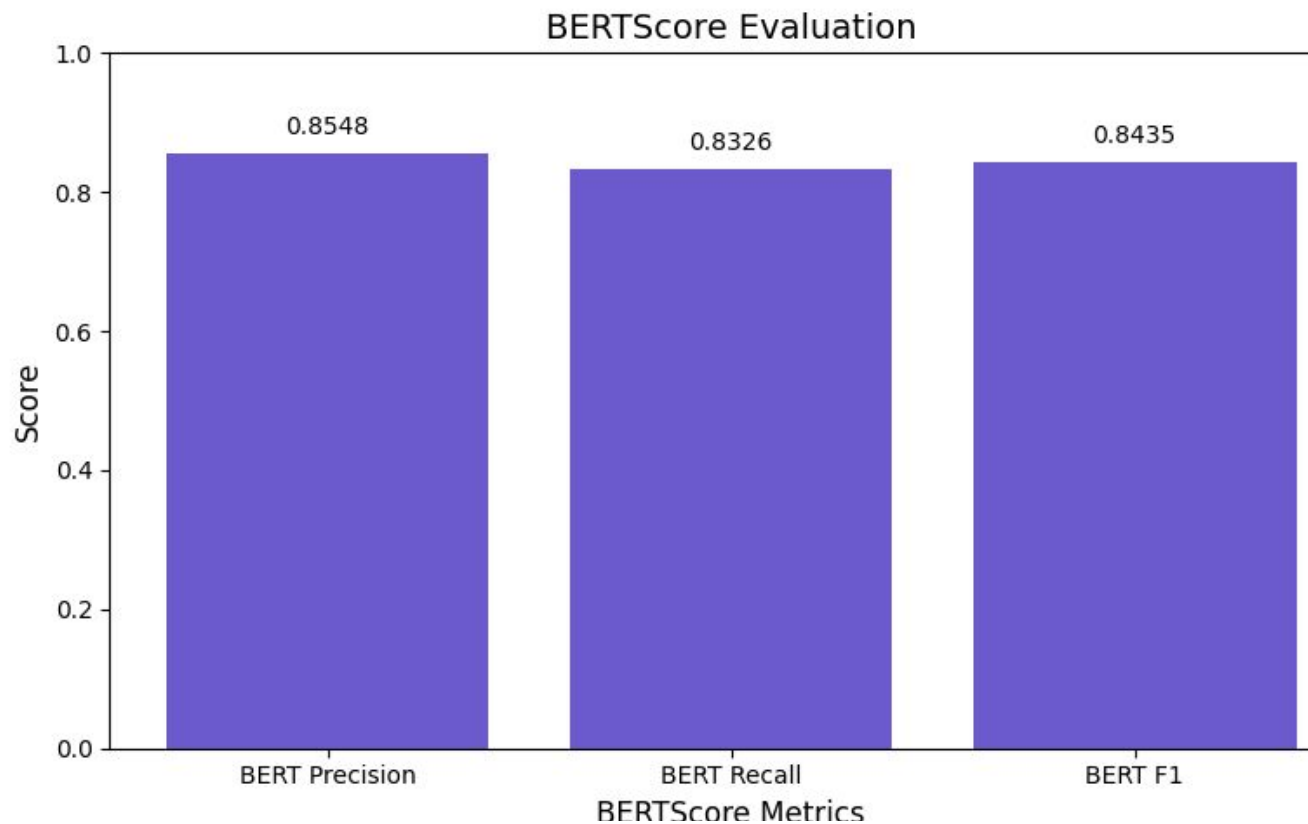


fig. (4) BLEU & METEOR evaluations of fine-tuned BART model



Output

1. Fine-Tuned BART model evaluation results



Output

2. Case Analysis

The screenshot displays a web application for legal document analysis. On the left, a document titled "Sanju @ Sanjay Singh Sengar vs State Of M.P on 1 May, 2002" is shown. The document includes case details such as "Equivalent citations: AIR 2002 SUPREME COURT 1998", "Bench: M.B. Shah, H.K. Sema", and "CASE NO.: Appeal (crl.) 572 of 2002". The document text includes the petitioner's name, the respondent's name, the date of judgment, and the bench. The judgment text is partially visible, starting with "SEMA, J Leave granted." and "Heard Mr. R.P. Gupta, learned Senior counsel on behalf of the appellant and Mr. B.S. Bantia, learned counsel on behalf of the respondent." The document is identified as "Page 1 of 5".

On the right, there is a section titled "Extracted Details" which lists the following information:

- Case Number: Appeal (crl.) 572 of 2002
- Date of Judgment: 1 May, 2002
- Appellant: Sanju @ Sanjay Singh Sengar
- Respondent: State Of M.P
- Bench: M.B. Shah, H.K. Sema
- Important Dates: 01 May, 2002

Below this, there is a section titled "Summary Options" with five buttons: "Extractive", "Abstractive", "Hybrid", "Concise", and "Precise". A "Generate Summary" button is located at the bottom of this section.

User uploads the legal document and gets the case details, as seen in 'Extracted Details' section, followed by summary options selection



Output

2. Case Analysis

Summary

History

it is stated that the marriage between the sister of the appellant and the deceased took place in 1993. it is also stated that immediately after marriage she was subjected to continuous ill treatment by the deceased and the family members forcing her to live separately along with her husband and children for about a year. on 25th july 1998 crucial date it is stated that the appellant visited the place of the parents of the deceased and pleaded with them that his sister should be rehabilitated in the matrimonial home and should not be physically ill treated or harassed. it is also alleged that on that day the appellant also said to have threatened the parents of the deceased that if they do not mend their behaviour towards his sister he would be compelled to resort to filing a complaint under section 498 a of the indian penal code to which the parents of the deceased expressed helplessness as the deceased chander bhushan had been living separately from them. it is further stated that on this story being narrated to the deceased by the mother of the deceased asking him to bring back his wife to avoid any police case against them the deceased went to the house of the parents of the appellant followed by a quarrel between them. this court was of the view that mere words uttered by the accused to the deceased to go and die were not even prima facie enough to instigate the deceased to commit suicide. reverting to the facts of the case both the courts below have erroneously accepted the prosecution story that the suicide by the deceased is the direct result of the quarrel that had taken place on 25th july 1998 wherein it is alleged that the appellant had used abusive language and had reportedly told the deceased to go and die . is annexed as annexure p 3 to this appeal and going through the statement we find that he has not stated that the deceased had told him that the appellant had asked him to go and die . it is common knowledge that the words uttered in a quarrel or in a spur of the moment cannot be taken to be uttered with mens rea. the deceased was found hanging on 27th july 1998. assuming that the deceased had taken the abusive language seriously he had enough time in between to think over and reflect and therefore it cannot be said that the abusive language which had been used by the appellant on 25th july 1998 drove the deceased to commit suicide. a plain reading of the suicide note would clearly show that the deceased was in great stress and depressed. she stated that the deceased always indulged in drinking wine and was not doing any work. it is in the statement of the wife that the deceased always remained in a drunken condition. it is a common knowledge that excessive drinking leads one to debauchery. it clearly appeared therefore that the deceased was a victim of his own conduct unconnected with the quarrel that had ensued on 25th july 1998 where the appellant is stated to have used abusive language. taking the totality of materials on record and facts and circumstances of the case into consideration it will lead to irresistible conclusion that it is the deceased and he alone and none else is responsible for his death.

Analyze

After selecting, desired summary options, user gets the summary by clicking on ‘Generate Summary’ button User can get detailed analysis by clicking ‘Analyze’



Output

2. Case Analysis

User gets detailed analysis of the case, like background, evidences, judgment, case outcome, etc.

Document Analysis:

Background ^

The appellant, Sanju Sanjay Singh Sengar, appealed against a charge framed against him for abetting the suicide of his brother-in-law under Section 306 of the Indian Penal Code (IPC). The suicide occurred on July 27, 1998, two days after a quarrel where the appellant allegedly abused and told the deceased to "go and die."

Evidence ^

The facts involved alleged ill-treatment of the appellant's sister, Neelam Sengar, by her husband (the deceased) and his family. The appellant visited the deceased's parents on July 25, 1998, threatening to file a police complaint under Section 498A of IPC if his sister was not treated well. Following this, a quarrel occurred between the appellant and the deceased where the deceased claimed he was threatened and abused by the appellant. Two days later, the deceased committed suicide and left a suicide note implicating the appellant.

Judgment v

Precedent Case ^

1. **Swamy Prahaladdas v. State of M.P. (1995 Supp. 3 SCC 438)**: Only uttering "go and die" isn't enough to instigate.
2. **Mahendra Singh v. State of M.P. (1995 Supp. 3 SCC 731)**: Harassment allegations alone aren't sufficient under Section 306 IPC.
3. **Ramesh Kumar v. State of Chhattisgarh (2001 9 SCC 618)**: Words spoken in anger without an intent for the consequences should not be seen as instigation.

Important Sections v

Case Outcome v



Conclusion

The project developed an efficient system for legal document summarization and analysis under the Bharatiya Nyaya Sanhita (BNS) framework. By fine-tuning InLegalBERT and BART models, the system ensures accurate section mapping, case analysis, and outcome prediction. It helps legal professionals transition smoothly from IPC to BNS and lays the foundation for future advancements in legal technology.



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